

An $O(\log n)$ -Competitive Posted-Price Algorithm for Online Matching on the Line

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Outline

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- **Motivation**

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- **Background 1: Posted-Price Algorithms**
 - Online Algorithms
 - Competitive Analysis
 - Online Metric Matching Problem
 - Posted-Price Metric Matching Problem

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- **Background 2: Online Matching on the Line**
 - Harmonic Algorithm
 - Doubled Harmonic Algorithm
 - **Modified Doubled Harmonic Algorithm**

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 - Harmonic Algorithm
 - Doubled Harmonic Algorithm
 - **Modified Doubled Harmonic Algorithm**
- **Conclusion**

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- **Goal:** Reduce time and fuel wasted
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- **Strategy:** Controlling **prices** to manage **demand**
- **NOTE:** Goal is **NOT** to **maximize profit!**



Online Algorithms

Online Algorithms

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- Example: Sorting!
 - **Online** algorithm: Insertion Sort
 - **Offline** algorithm: Selection Sort

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 - ON is α -competitive if $ON(I) \leq \alpha \cdot OPT(I)$ for all instances I

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- Minimize the cost of matching the Requests (Cars) to the Servers (Parking Spots)!

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Important Note: Online algorithm **tells** the cars where to park!

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- **High-Level:** Online algorithm controls **incentives (\$\$\$)** of cars, but not their **decisions!**

Posted-Price Metric Matching Problem

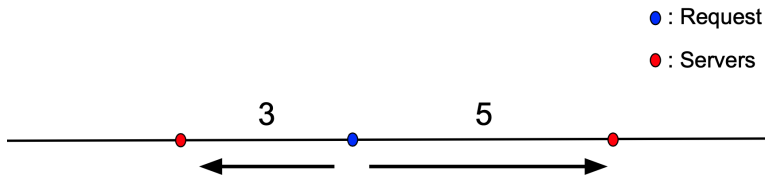
- **High-Level:** Online algorithm controls **incentives (\$\$\$)** of cars, but not their **decisions!**
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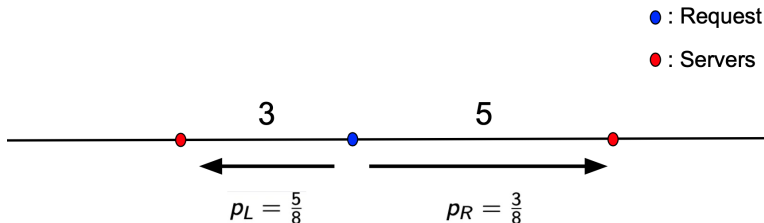
- **High-Level:** Online algorithm controls **incentives (\$\$\$)** of cars, but not their **decisions!**
- **Server Prices:** Online algorithm sets server prices (parking meters) on available servers (parking spots)
- **Agent Decision:** Requests (cars) will move **selfishly**.

Agent Minimizes: Distance + Price

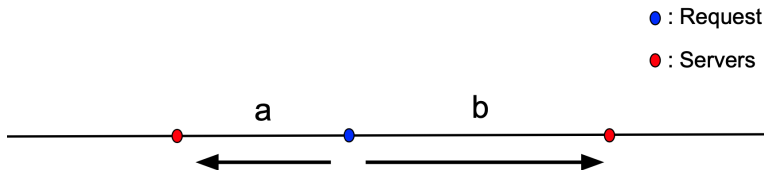
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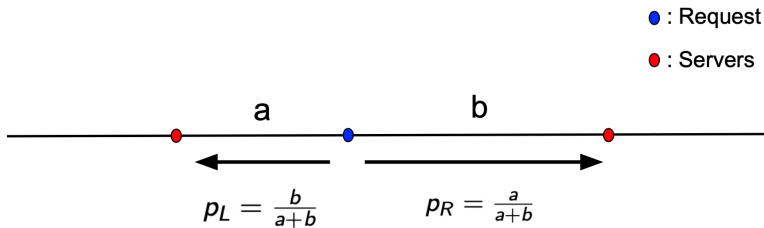
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**Implementable as
Posted-Price Algorithm**



$O(\log \Delta)$ -Competitive



Δ : Aspect ratio of instance

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Doubled Harmonic Algorithm

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- **Imaginary Servers:** Maintains “imaginary servers” it wishes were available
- **Guess-and-Double:** Once OPT's current cost exceeds Z , double Z ! Reshuffle the imaginary servers!

Doubled Harmonic Algorithm

**Not implementable as
Posted-Price Algorithm**



$O(\log n)$ -Competitive



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Two Major Problems

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- 1 Imaginary Servers are **not coordinated** with the Available Servers in a “posted-price” way.

Doubled Harmonic Algorithm

Two Major Problems

- ① Imaginary Servers are **not coordinated** with the Available Servers in a “posted-price” way.
- ② **Reshuffling** the Imaginary Servers is not posted-price implementable.

Modified Doubled Harmonic Algorithm

Modified Doubled Harmonic Algorithm

Two Major Solutions

Modified Doubled Harmonic Algorithm

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- 1 We properly **coordinated** the imaginary servers with the available servers.

Modified Doubled Harmonic Algorithm

Two Major Solutions

- ① We properly **coordinated** the imaginary servers with the available servers.
- ② We **delayed** the reshuffling procedure and showed the cost was not significantly damaged.

Modified Doubled Harmonic Algorithm

**Implementable as
Posted-Price Algorithm**



$O(\log n)$ -Competitive



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- San Francisco is **not** a **line metric!**
- $O(\log \Delta)$ to $O(\log n)$ improvement is not practically significant!

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- **Major Goal:** $\text{polylog}(n)$ -Competitive for **Tree Metrics**
 - **Past Work 1:** $\text{polylog}(n)$ -Comp. for “Tree Search” Instances
 - **Past Work 2:** $\text{polylog}(n)$ -Comp. for “Spider” Instances

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- **Our Goal:** Implement the ubiquitous **guess-and-double** technique in online algorithms using **prices**.
- **Hope:** Extend this work to **tree metrics!**

Questions?

